

Neel Panging

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EXPERIENCE

realtor.com

August 2025 - Present

Senior Machine Learning Engineer

Austin, TX (Hybrid)

- Engineered production-grade, personalized search-ranking pipelines (XGBoost, Metaflow), optimizing parallel ETL workflows to reduce runtime by 81% while strictly maintaining output parity.
- Optimized model deployment infrastructure and query serving, automating Elasticsearch feature pruning to eliminate 65% of latency-inducing scripts while accelerating release candidate evaluation.
- Spearheaded an automated ML research program and deployed robust A/B tests for lead-sensitive ranking objectives, driving a 10%+ wNDCG improvement and a 7% lift in top-10 target conversions.
- Architected a Snowflake and Streamlit model-evaluation platform deployed via Kubernetes, enabling rapid, side-by-side ranking validation across millions of listings without full train-and-deploy cycles.
- Built comprehensive ML observability and monitoring systems, centralizing deployment metadata and implementing automated anomaly detection with Slack alerting for click/view tracking.

Transfix - Digital Freight Marketplace

July 2023 - August 2025

Data Scientist II

Austin, TX (Remote)

- Increased pricing accuracy by developing a novel bottom-up forecasting model predicting short-term trucking rates for 18,000+ market lanes. Supported decisions via statistical analysis of freight market rates/volatility and model backtests for accuracy validation.
- Boosted model reliability and decreased win-rate bias by over 80% in Transfix's core freight pricing algorithm, directly enhancing pricing strategy and decision-making.
- Built ETL pipelines (Snowflake, DBT) and Streamlit apps for forecast visualization and model explainability.
- Improved user engagement by 45% (reducing empty recommendations) via enhancements to Transfix's freight recommender platform. Revised algorithms to serve new/infrequent users beyond 60-day activity thresholds.
- Optimized MLOps: Deployed models on AWS SageMaker and Argo. Managed workflows via Airflow DAGs and standardized deployments using Docker, GitHub, and CI/CD pipelines.

Dell Technologies

June 2020 - July 2023

Data Science Intern - Data Scientist I

Austin, TX

- Saved Dell approximately \$1.2 million annually in compute costs by developing an ensemble classification model (Catboost + Autoregressor) to predict server backup failures from server logs, reducing unexpected server downtimes and optimizing resource allocation.
- Enhanced customer service efficiency by 18% by implementing a BERT model to automate the summarization and classification of customer service cases, saving support agents dozens of hours each week.
- Experimented with fine-tuning and prompt engineering on various LLMs, including BART and GPT.
- Optimized agent workload allocation and improved service efficiency by 7% by forecasting product-level service requests using Prophet models, enabling demand planners to balance agent workloads effectively and reducing customer wait times.

EDUCATION

Georgia Institute of Technology - MS Computer Science | Machine Learning Specialization

2024

The University of Texas at Austin - BS Applied Mathematics | Minors in Computer Science and Statistics

2022